XuannvEarth: Toward China-Scale Monthly Geospatial Embeddings for Reusable Sparse-Label Mapping

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Abstract

China's urban and regional governance increasingly requires monthly, fine-grained maps of buildings, roads, water bodies, green spaces, schools, sports facilities, construction activities, and other urban objects. However, dense task-specific annotations are expensive to update, and conventional mapping models are usually trained separately for each target category. We present XuannvEarth, a monthly multimodal geospatial embedding framework that converts heterogeneous Earth observation data into reusable dense representations for sparse-label urban mapping. The central idea is to generate the embedding once and reuse it across many downstream tasks by attaching simple linear, MLP, convolutional, or lightweight segmentation heads, instead of retraining a separate heavy model for every target. In this study, we validate the framework over Haidian District, Beijing, a complex urban region with dense buildings, road networks, water bodies, parks, campuses, and sports facilities. XuannvEarth integrates Sentinel-2 optical imagery, Sentinel-1 SAR, Landsat observations, high-resolution optical imagery, high-resolution SAR imagery, and OpenStreetMap-derived weak semantic labels to produce 10 m, 64-dimensional monthly embedding maps.

XuannvEarth is trained to preserve multi-sensor appearance, high-resolution spatial detail, monthly surface state, and weak urban semantics in a unified representation. Its objective uses reconstruction-and-weak-supervision learning with masked-source training and context-aware patch cropping, encouraging the embedding to learn stable surface representations rather than only task-specific classification boundaries. After pre-training, the encoder is frozen. Each downstream task only reads the generated embedding map and trains a small task head, making the workflow suitable for rapid category expansion when only a few labeled patches are available. All evaluations use a unified protocol with fixed train/validation/test splits, validation-selected thresholds, and matched label budgets.

Experiments demonstrate that one monthly embedding product can support multiple downstream mapping tasks. For 5-shot building, road, and water extraction, we compare XuannvEarth against task-specific baselines trained directly from April 2026 multisensor features, including Sentinel-2, Sentinel-1, Landsat, high-resolution optical imagery, and high-resolution SAR features. The baselines use the same data splits, the same number of labeled patches, the same validation-threshold selection rule, and the same candidate heads, including Conv3x3, MLP, U-Net, and DeepLab-lite. The reported gains of 9.8%, 18.3%, and 41.5% in F1 are relative to the best raw-feature baseline selected separately for each task under this identical protocol. In the 50-shot six-task benchmark, XuannvEarth embeddings with lightweight or spatial heads achieve an average F1 of 0.552 and outperform raw-feature U-Net, DeepLab-lite, or SegFormer-lite baselines on five of six urban mapping tasks.

The project is designed as an open GeoAI workflow rather than a closed single-task model. We will release trained weights, monthly embeddings, training configurations, data documentation, downstream evaluation scripts, visualization examples, and reproducible ModelScope notebooks. Looking forward, we aim to scale XuannvEarth from regional validation to a China-oriented monthly geospatial embedding foundation, enabling reusable sparse-label mapping, query-by-example retrieval, and rapid urban semantic analysis across diverse Chinese landscapes.

Index Terms--geospatial embeddings, remote sensing foundation models, sparse-label mapping, urban GeoAI, multimodal representation learning, ModelScope